

OP-AMP

An OP-AMP is a high gain, direct-coupled, differential amplifier.

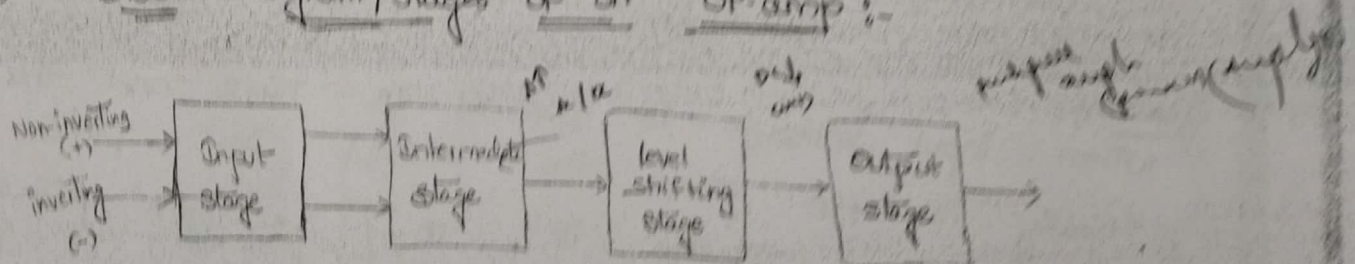
Direct-coupled: An OP-AMP is a direct couple because, the output of the one OP-AMP can be input to another OP-AMP.

High-gain: It is called high-gain differential amplifier because the difference of the two input is amplified.

OP-AMP stands for operational amplifier because it is used for mathematical operations such as addition, subtraction, multiplication, division, differentiation and integration. It is designed with various types of active devices and it provides low cost and high performance. It is used in signal processing (Stereo equipment and medical cardiology) etc.

Stages

Block diagram/stages of an OP-amp:-



(a) The input stage consists of dual input - balanced output differential amplifier. It provides high voltage gain, high input impedance.

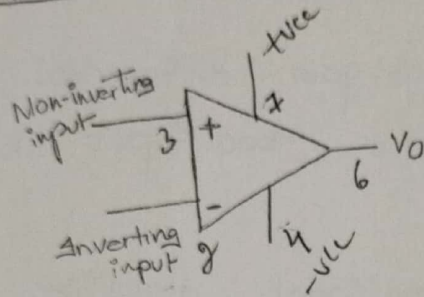
(b) This stage consists of dual input - unbalanced output differential amplifier. It provides amplification. The output of the intermediate stage consists of AC

as well as DC signal.

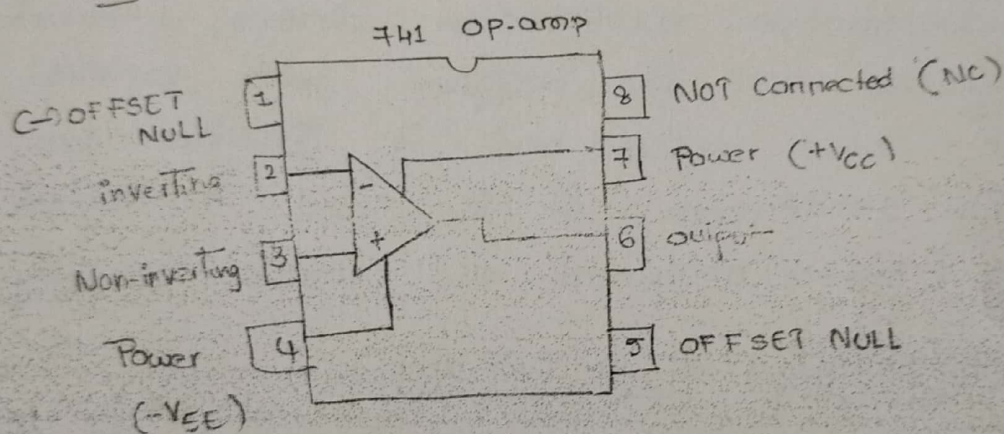
(c) This stage consist of emitter follower with constant current. Due to unbalanced output it shift the dc level.

(d) This stage consist of complimentary symmetry Push-pull amplifier. At raising the voltage swing and current supply capability of OP-AMP

Symbols of an op-amp :-



Pin diagram :-



Inverting:
When input voltage is applied to pin 2 and the output in pin 6
Polarity is +ve then the output in pin 6
having the polarity is -ve [i.e. 180° phase shift]

Non-inverting:
When the input voltage is applied to pin 3 and the output in pin 6
Polarity is +ve then the output in pin 6
having +ve polarity.

Characteristics of op-amp: [741 IC]

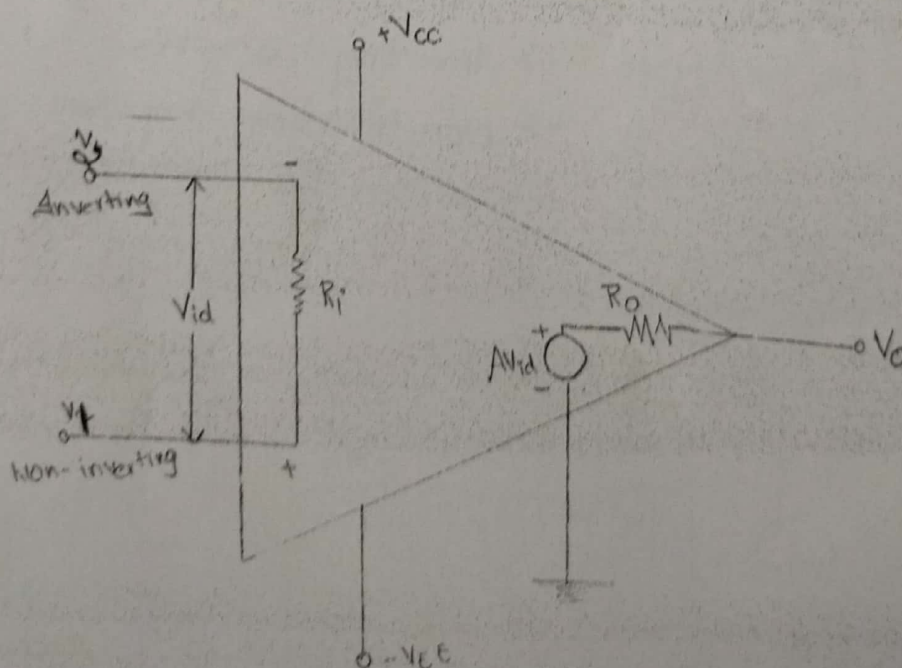
- High voltage gain ($10^4 - 10^5$)
- High input impedance (1 to $2 \text{ M}\Omega$)
- Low output impedance ($\sim 100 \Omega$)
- Large bandwidth
- Large CMRR ($70 - 80 \text{ dB}$)
↓
(Common Mode Rejection Ratio)
- Large Slew rate ($\sim 0.5 \text{ V}/\mu\text{s}$)

$$SR = \left(\frac{dV_o}{dt} \right)_{\max}$$
$$CMRR = \frac{|A_d|}{|A_c|}$$

Ideal characteristics of op-amp:

- High voltage gain ($A = \infty$)
- High input impedance ($R_i = \infty$)
- Low Output Impedance ($R_o = 0$)
- Large bandwidth ($B.W = \infty$)
- Large CMRR ($CMRR = \infty$)
- Large Slew rate (Slew rate $= \infty$)

Equivalent circuit of op-amp:



$A = \text{gain}$

$V_{id} = \text{difference input voltage}$

$$= V_1 - V_2$$

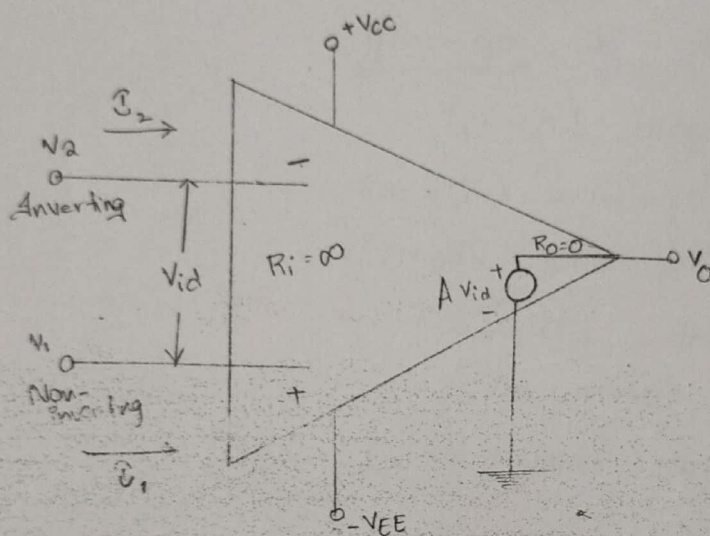
$V_O = \text{output voltage}$

$$V_O = A \cdot V_{id}$$

$$= A \cdot (V_1 - V_2)$$

The output voltage is directly proportional to the difference of two input voltages.

Ideal equivalent circuit of op-amp:



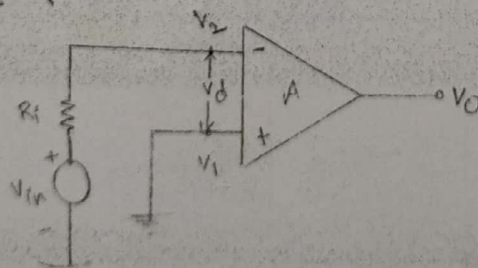
$$I_2 = \frac{V_2}{R_i} = \frac{V_2}{\infty} = 0$$

$$I_1 = \frac{V_1}{R_i} = \frac{V_1}{\infty} = 0$$

a) Open-loop Configuration [No feedback from o/p to i/p]

- It is of 3 types $\rightarrow$ Inverting amplifier
- $\rightarrow$ Non-inverting amplifier
- $\rightarrow$ Differential amplifier

$\rightarrow$ Inverting amplifier:



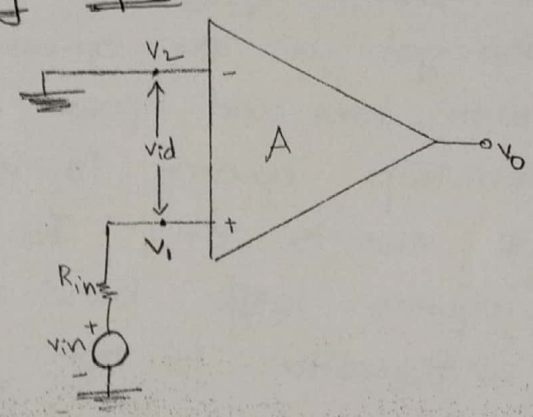
The input signal is applied to the inverting terminal and non-inverting terminal is connected to the ground

$$\begin{aligned} V_o &= A \cdot V_{id} \\ &= A (V_1 - V_2) \\ &= -A V_2 \end{aligned}$$

$$\boxed{V_o = -A V_{in}}$$

In the above equation, the output voltage is 180° (out of phase) w.r.t input.

→ Non-inverting amplifier:-



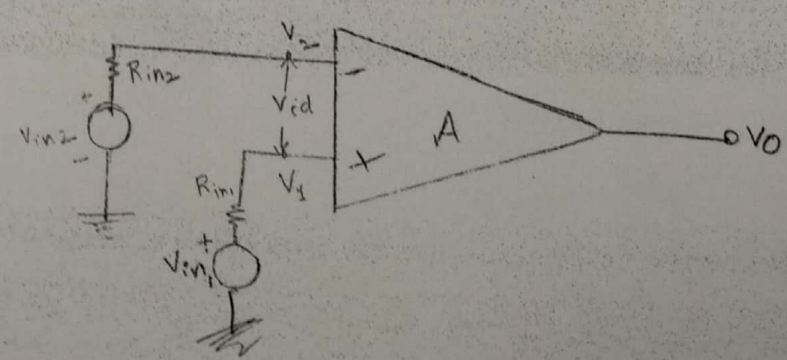
The input signal is applied to non-inverting terminal and inverting terminal is connected to the ground.

$$\begin{aligned} V_o &= A \cdot V_{id} \\ &= A (V_1 - V_2) = A V_1 \end{aligned}$$

$$\boxed{V_o = A V_{in}}$$

In the above eqn, the output voltage is in phase w.r.t input.

→ Differential amplifier:-



The inputs are applied to both inverting & non-inverting terminals.

$$V_o = A V_d$$

$$= A (V_1 - V_2)$$

$$= A (V_{in1} - V_{in2})$$

The output voltage is directly proportional to difference of two input voltages. So, it is called differential amplifier.

13/10/14

Limitation of open-loop configuration:

The open-loop gain of the op-amp is not constant it changes with temp and power supply. The band width of open-loop op-amp. is very strong. The band width of $\mu 741$ is 5 Hz. For amplification purpose, it requires large band width. Hence, open loop configuration is not used for linear amplification.

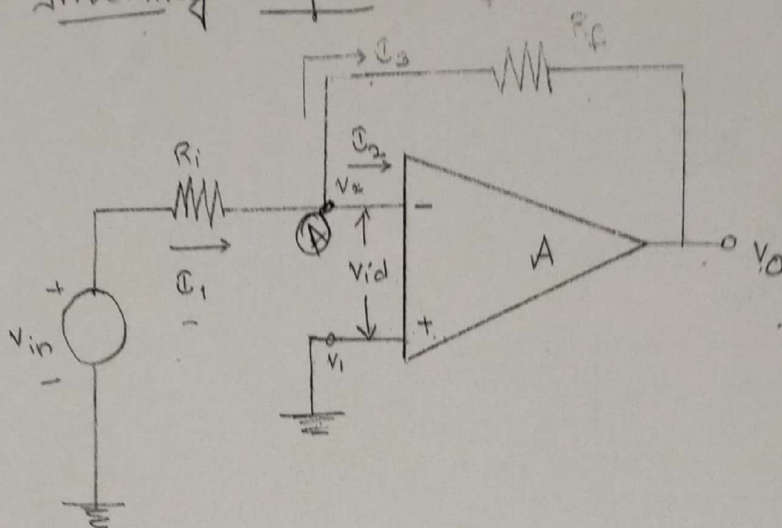
> closed-loop configuration:

The op-amp can effectively be used for linear amplification by providing feed back from ϕ/p to ϕ/p .

AF signal feedback is out of phase by 180° w.r.t input, then the feedback is called negative feedback / degenerative feedback.

AF signal feedback is in phase with ϕ/p , then the feedback is called positive feedback / regenerative feedback.

→ Inverting Amplifier :-



In the above fig, it shows inverting amplifier with closed loop configuration

R_i : Input resistor

R_f : feedback resistor

$$V_o = A \cdot V_{id}$$

$$= A (V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{V_o}{A} = \frac{V_o}{\infty} = 0 \rightarrow \textcircled{1}$$

$\therefore A$ = open-loop gain.

In the equⁿ $\textcircled{1}$, as the difference of two i/p voltages is equal to zero, is called virtual short ckt cond'n

From equⁿ $\textcircled{1}$,

$$V_1 - V_2 = 0$$

$$V_1 = V_2 = 0 \rightarrow \textcircled{2}$$

as the two voltage is equal to zero, is called virtual ground cond'n.

Apply kcl at point A,

$$I_1 = I_2 + I_3$$

$\therefore I_2 = 0$ (for ideal op-amp)

$$\Rightarrow I_1 = I_3$$

$$\Rightarrow \frac{V_{in} - V_2}{R_1} = \frac{V_2 - V_0}{R_f}$$

$$\Rightarrow \frac{V_{in}}{R_1} = -\frac{V_0}{R_f} \quad (V_2 = 0)$$

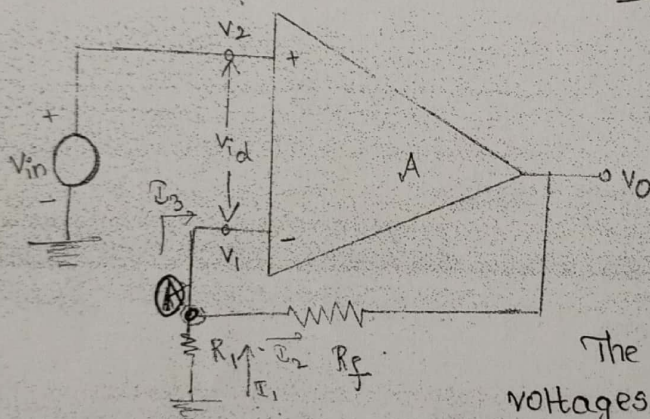
$$\Rightarrow V_0 = -\frac{R_f}{R_1} \cdot V_{in}$$

$$\Rightarrow \frac{V_0}{V_{in}} = A_f = -\frac{R_f}{R_1}$$

$\therefore A_f$ is called feedback gain.

10/10/17

$\Rightarrow$ Non-inverting Amplifier:



for open loop,

$$V_0 = A \cdot V_d$$

$$= A(V_2 - V_1)$$

$$\Rightarrow V_2 - V_1 = \frac{V_0}{A} = \frac{V_0}{\infty} = 0$$

The difference of two input voltages is equal to zero is called virtual short ckt condition

$$\Rightarrow V_2 - V_1 = 0$$

$$V_2 = V_1 = V_{in}$$

for closed loop,

Apply KCL at A

$$I_1 = I_2 + I_3$$

$\therefore I_3 = 0$ for ideal op-amp

$$I_1 = I_2$$

$$\Rightarrow \frac{0 - V_k}{R_1} = \frac{V_i - V_o}{R_f}$$

$$-\frac{V_{in}}{R_1} = \frac{V_{in} - V_o}{R_f}$$

$$\Rightarrow -\frac{V_{in}}{R_1} = \frac{V_{in}}{R_f} - \frac{V_o}{R_f}$$

$$\Rightarrow -\frac{V_{in}}{R_1} - \frac{V_{in}}{R_f} = -\frac{V_o}{R_f}$$

$$\Rightarrow V_{in} \left(\frac{1}{R_1} + \frac{1}{R_f} \right) = \frac{V_o}{R_f}$$

$$V_{in} \left(\frac{R_f + R_1}{R_1 R_f} \right) = \frac{V_o}{R_f}$$

$$V_o = V_{in} \left(\frac{R_1 + R_f}{R_1} \right) \times \frac{R_f}{R_f}$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) \cdot V_{in}$$

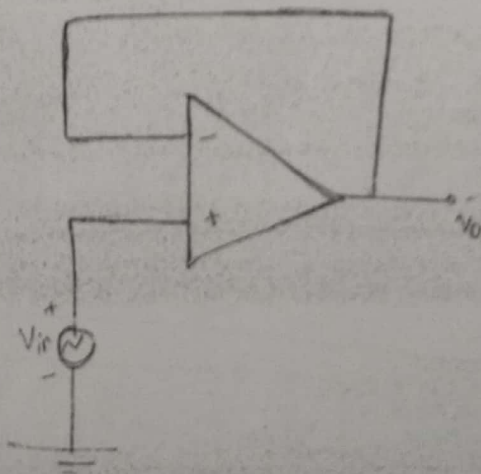
$$\Rightarrow \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_1}$$

$$A_f = 1 + \frac{R_f}{R_1}$$

17/10/17

Op-amp Applications:

→ Voltage follower Buffer:



If $R_1 = \infty$ & $R_f = 0$ in non-inverting amp, then the amp. is called unity-gain amp.

$$A = 1 + \frac{R_f}{R_1}$$

$$A_f = 1$$

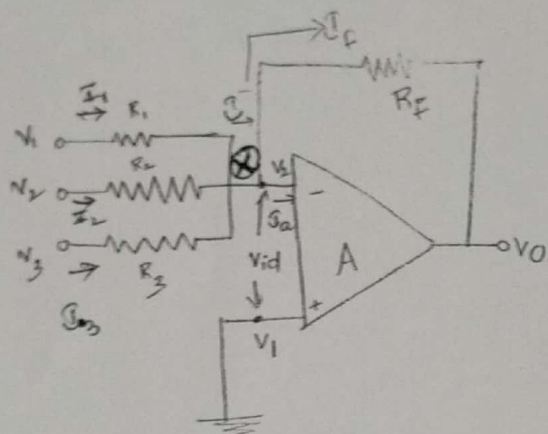
$$\text{As } A_f = 1$$

$$\Rightarrow \frac{V_o}{V_{in}} = 1$$

$$V_o = V_{in}$$

Since the output voltage follows the input voltage, it is called voltage follower.

Q2 → Adder or Summing Amplifier:



linear addition,

$$V_0 = A \cdot V_{id} = A(V_1 - V_2)$$

$$\Rightarrow (V_1 - V_2 = \frac{V_0}{A}) = \frac{V_0}{\infty} = 0$$

$$V_1 - V_2 = 0$$

The above circuit shows is Adder/Summing operation.
The output of the circuit is linear addition of input signal.

$$V_0 = A \cdot V_{id} = A(V_1 - V_2)$$

$$V_1 - V_2 = \frac{V_0}{\infty} \Rightarrow V_1 - V_2 = 0$$

Applying KCL at $-$

$$I = I_a + I_f$$

$$I = I_f \quad (\because I_a = 0 \text{ for ideal})$$

$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1 - V_2}{R_1} + \frac{V_2 - V_2}{R_2} + \frac{V_3 - V_2}{R_3} = \frac{V_2 - V_0}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{V_0}{R_f}$$

$$\Rightarrow V_0 = \left(\left(1 + \frac{R_f}{R_1}\right) V_1 + \left(1 + \frac{R_f}{R_2}\right) V_2 + \left(1 + \frac{R_f}{R_3}\right) V_3 \right)$$

AF $R_1 = R_2 = R_3 = R_f$

$$V_0 = -(V_1 + V_2 + V_3)$$

~~$$V_0 = \left(\left(1 + \frac{R_f}{R_1}\right) V_1 + \left(1 + \frac{R_f}{R_2}\right) V_2 + \left(1 + \frac{R_f}{R_3}\right) V_3 \right)$$~~

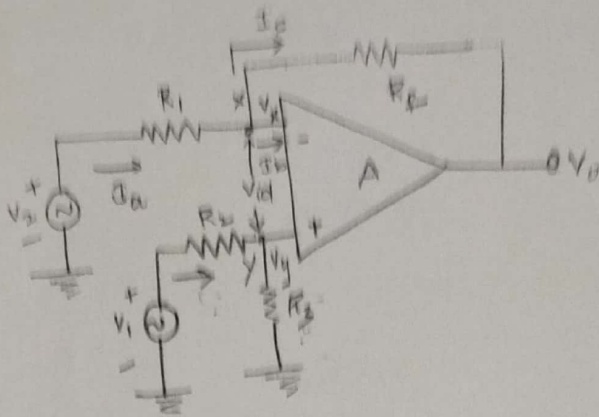
$$V_0 = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

Subtractor or differential amplifier

[Differential amplifier]

A single op amp differential amplifier

6



The above circuit diagram is used as subtractor, to analysis the operation, assume all the resistors.

$$V_0 = A \cdot V_{id} = A (V_y - V_x)$$

$$V_y - V_x = \frac{V_0}{A} = \frac{V_0}{\infty} = 0$$

$$\Rightarrow \boxed{V_y = V_x}$$

$$V_y = \frac{V_1 \times R_3}{R_2 + R_3}$$

Apply KCL at x, $\Rightarrow I_a = I_b + I_c$

$$\Rightarrow I_a = I_c$$

$$\frac{V_2 - V_x}{R_1} = \frac{V_x - V_0}{R_F}$$

$$\Rightarrow \frac{V_2}{R_1} - \frac{V_x}{R_1} = \frac{V_x}{R_F} - \frac{V_0}{R_F}$$

$$\Rightarrow \frac{V_2}{R_1} - \frac{V_x}{R_1} - \frac{V_x}{R_F} = -\frac{V_0}{R_F}$$

$$\Rightarrow V_0 = -\frac{R_F V_2 + R_F V_x \left(\frac{1}{R_1} + \frac{1}{R_F} \right)}{R_1}$$

$$= -\frac{R_F}{R_1} V_2 + R_F V_x \left(\frac{1}{R_1} + \frac{1}{R_F} \right)$$

$$= -\frac{R_F}{R_1} V_2 + R_F \times \frac{V_1 R_3}{R_2 + R_3} \left(\frac{1}{R_1} + \frac{1}{R_F} \right)$$

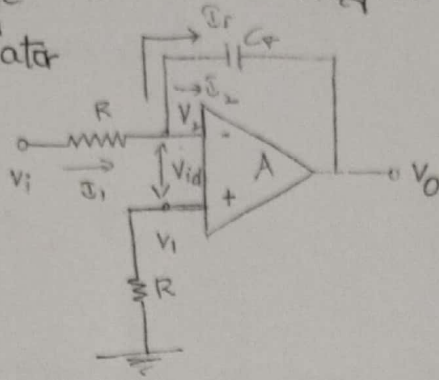
$$= -\frac{R_F}{R_1} V_2 + \frac{R \times R}{R + R} \cdot V_1 \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$= -V_2 + V_1$$

$$\boxed{V_0 = V_1 - V_2}$$

18/10/11

Integrator: The circuit in which the output of the voltage is the integral of input voltage is called integrator



$$V_o = A \cdot V_{id}$$

$$= A \cdot (V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{V_o}{A} = \frac{V_o}{\infty} = 0$$

$$V_1 = V_2 = 0$$

Apply KCL at point A,

$$I_1 = I_2 + I_f$$

$$\Rightarrow I_1 = I_f \quad (\because I_2 = 0)$$

$$\Rightarrow \frac{V_i - V_2}{R_L} = C_f \cdot \frac{dV_o}{dt}$$

$$= C_f \cdot \frac{d}{dt} (V_2 - V_o)$$

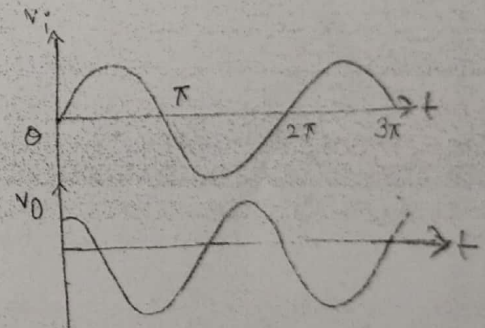
$$\Rightarrow \frac{V_i}{R} = C_f \frac{d}{dt} (-V_o)$$

Integrate both sides

$$\Rightarrow \int \frac{V_i}{R} dt = -C_f V_o$$

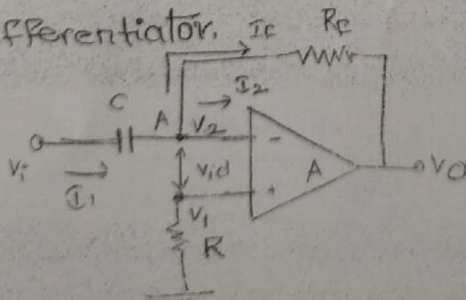
$$\Rightarrow \boxed{V_o = -\frac{1}{R C_f} \int V_i dt}$$

output wave form



Differentiator:

The circuit in which the output of the voltage is the differential of input voltage is called differentiator.



$$V_o = A \cdot V_{id}$$

$$= A \cdot (V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{V_o}{A} = \frac{V_o}{\infty} = 0$$

Apply KCL,

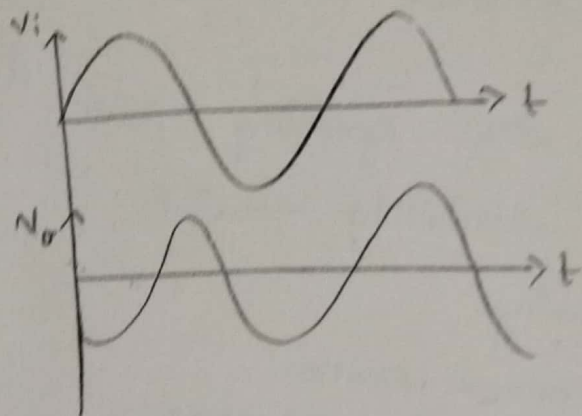
$$I_1 = I_2 + I_f$$

$$I_1 = I_f$$

$$C \frac{dv_o}{dt} = \frac{v_a - v_o}{R_f} \Rightarrow C \frac{d}{dt} (v_i - v_2) = \frac{v_2 - v_o}{R_f}$$

$$C \frac{d}{dt} (v_i) = - \frac{v_o}{R_f}$$

$$v_o = -R_f C \frac{d}{dt} (v_i)$$



1. Why BJT is called Bipolar Junction Transistor?
2. How BJT acts as an amplifier and a switch?
3. Explain the working of N-P-N & P-N-P transistor?
4. Explain common base and common emitter configuration with α , β & δ , Numericals related to α , β & δ .
5. Explain input, output characteristics of common base and common emitter.
6. Explain transistor load line with operating point.
7. Explain any biasing with stability factor.
8. Numericals on transistor biasing.
9. Small Signal analysis of common emitter.
10. What is MOSFET and types of MOSFET.
11. Operation of depletion and Enhancement of MOSFET.
12. Biasing on depletion & Enhancement MOSFET (Numericals)

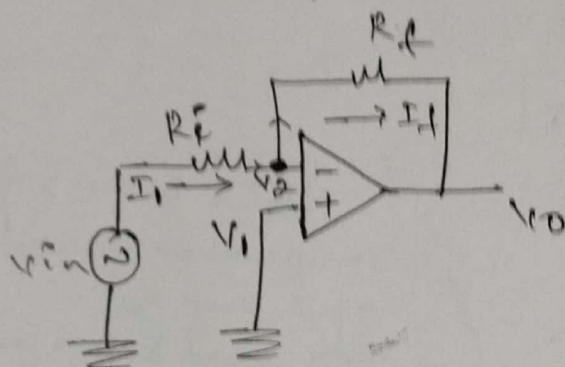
Effect of finite open loop gain (ov) (8)
 Effect of open-loop gain on closed loop gain
 of an amplifier.

$$\text{Theoretical Gain } A_{cl} = \frac{-R_f}{R_i}$$

a) Inverting (practical)

Generally

$$A = \frac{V_o}{V_{in}}$$



$$V_o = A_{OL}(V_1 - V_2)$$

open loop gain.

Since $V_1 = 0$

$$V_2 = -\frac{V_o}{A_{OL}}$$

Apply KCL @ V_2 pt

$$I_i = I_f$$

$$\frac{V_{in} - V_2}{R_i} = \frac{V_2 - V_o}{R_f}$$

$$\frac{V_{in}}{R_i} + \frac{V_o}{R_f} = V_2 \left[\frac{1}{R_f} + \frac{1}{R_i} \right]$$

on replacing V_2

$$\frac{V_{in}}{R_i} + \frac{V_o}{R_f} = \frac{-V_o}{A_{OL}} \left[\frac{1}{R_f} + \frac{1}{R_i} \right]$$

$$\frac{V_o}{R_f} + \frac{V_o}{A_{OL}} \left[\frac{1}{R_f} + \frac{1}{R_i} \right] = -\frac{V_{in}}{R_i}$$

$$V_o \left[\frac{1}{R_f} + \frac{1}{A_{OL}} \left(\frac{1}{R_f} + \frac{1}{R_i} \right) \right] = -\frac{V_{in}}{R_i}$$

$$\frac{V_o}{V_{in}} = \frac{-1}{\frac{R_i}{R_f} + \frac{1}{A_{OL}} \left(\frac{R_i}{R_f} + 1 \right)}$$

$$\frac{V_o}{V_{in}} = \frac{-R_f}{R_i + \frac{1}{A_{OL}}(R_i + R_f)}$$

÷ by R_i

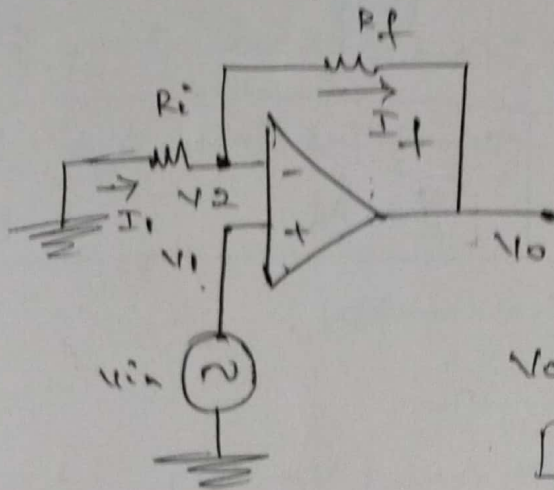
$$\frac{V_o}{V_{in}} = \frac{-R_f/R_i}{1 + \frac{1}{A_{OL}} \left(1 + \frac{R_f}{R_i} \right)}$$

∴

$$A_{CL} = \frac{V_o}{V_{in}} = \frac{-R_f/R_i}{1 + \frac{1}{A_{OL}} \left(1 + \frac{R_f}{R_i} \right)}$$

A_{CL} = closed loop gain.

b) Non-inverting (op-amp)



Theoretical Gain Ideal

$$\Rightarrow A_{CL} = 1 + \frac{R_f}{R_i}$$

$$V_o = A_{OL} (V_1 - V_2) \quad \text{--- (1)}$$

$$V_1 = v_{in}$$

Apply KCL @ V_2 pt

$$I_1 = I_f$$

$$\frac{0 - V_2}{R_i} = \frac{V_2 - V_o}{R_f}$$

$$V_2 \left[\frac{1}{R_i} + \frac{1}{R_f} \right] = \frac{V_o}{R_f}$$

$$V_2 = \frac{V_o}{\left[1 + \frac{R_f}{R_i} \right]} \quad \text{--- (2)}$$

sub (2) in (1)

$$V_o = A_{OL} \left[v_{in} - \frac{V_o}{1 + \frac{R_f}{R_i}} \right]$$

$$V_o \left[1 + \frac{A_{OL}}{1 + \frac{R_f}{R_i}} \right] = A_{OL} v_{in}$$

$$\frac{V_o}{v_{in}} = \frac{A_{OL}}{1 + \frac{A_{OL}}{1 + \frac{R_f}{R_i}}}$$

Take LCM

NOTE: Because of Temperature, pressure open loop gain may vary so we are making it finite

$$\frac{V_o}{v_{in}} = \frac{A_{OL}}{\left(1 + \frac{R_f}{R_i} \right) + A_{OL}}$$

$$= \frac{\left(1 + \frac{R_f}{R_i} \right) A_{OL}}{\left(1 + \frac{R_f}{R_i} \right) + A_{OL}}$$

$$A_{CL} = \frac{\left(1 + \frac{R_f}{R_i} \right)}{\left(1 + \frac{R_f}{R_i} \right) \frac{1}{A_{OL}} + 1}$$

$\therefore$ closed loop gain is

$$A_{CL} = \frac{V_o}{v_{in}} = \frac{1 + \frac{R_f}{R_i}}{1 + \frac{1}{A_{OL}} \left[1 + \frac{R_f}{R_i} \right]}$$